

NAME: NGENO KIBET CALEB

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UNIT CODE: BIT 2324

UNIT NAME: Geographical Information Systems

FEB-MAY 2026 ASSIGNMENT

1. Describe the stages of creating digital datasets.

a. **Planning and Design:**

Defining the problem or research question, establishing the scope of the dataset, and clarifying its purpose. This foundational step guides all subsequent activities by setting clear objectives.

b. **Establishing Criteria:**

Setting standards and protocols for data collection, labeling, annotation, and cleaning. These criteria ensure consistency, accuracy, and relevance throughout the dataset creation process.

c. **Collation:**

Gather raw data from various sources or generate new data through experiments or simulations. This stage involves importing data into digital formats and structuring it appropriately.

d. **Validation – Quality Assurance:**

Performing checks to verify data accuracy, completeness, and integrity. This includes data validation procedures to maintain high-quality standards. Decide whether the output solves or answers the problem/question. If not, then refine the problem and start the process again.

e. **Analysis:**

Examine and interpret the data to extract meaningful insights, identify patterns, or prepare it for specific applications such as machine learning or reporting.

f. **Storage and Organization:**

Store the dataset systematically using suitable formats and structures, ensuring easy access, retrieval, and management of data assets.

g. **Documentation – Metadata:**

Create comprehensive documentation detailing data sources, collection methods, preprocessing steps, and structure. Metadata enhances transparency and reproducibility.

h. **Maintenance – Updating:**

Regularly update and revise the dataset to incorporate new data, correct errors, or adapt to changing requirements, ensuring ongoing relevance and accuracy.

i. **Distribution:**

Sharing the final dataset with users via repositories, platforms, or direct dissemination, facilitating access and use by stakeholders or the community.

2. The creation of a clean digital database is the most important and time-consuming task upon which the usefulness of the GIS depends.

Reference

- Suresh, H., & Gutttag, J. V. (2019). A framework for understanding sources of harm throughout the machine learning life cycle. <https://arxiv.org/abs/1901.10002>
- Google. (n.d.). Machine learning data preparation, from <https://developers.google.com/machine-learning/data-preparation>
- Victoria, B., & Conway, D. (n.d.). Data Version Control (DVC) documentation. Retrieved April 14, 2026, from <https://dvc.org/doc>
- ISO 19115 – Geographic Information and ISO 55013 – Asset Management.